

# Nearest Neighbor Reasoning

Project: Transparent Book Recommender

## 1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Similarity by Selected Traits

Without opening the old lesson, explain one key idea from Similarity by Selected Traits. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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### Sampling and Representation

Without opening the old lesson, explain one key idea from Sampling and Representation. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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### Overfitting Intuition

Without opening the old lesson, explain one key idea from Overfitting Intuition. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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## 2. Practise with fading help

Use W2: Apply neighbour and tie rules for Nearest neighbours and distances. Record the answer once; the lesson and workbook refer to the same attempt.

**Given inputs · Procedure I chose · Middle steps · Result and check**

**Given inputs:** \_\_\_\_\_

**Procedure I chose:** \_\_\_\_\_

**Middle steps:** \_\_\_\_\_

**Result and check:** \_\_\_\_\_

\_\_\_\_\_

**The step I completed without help, and the hint I needed**

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## 3. Individual reasoning check

Use the worked case for Nearest neighbours and distances. Proposed explanation: “A negative difference is a negative distance”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

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## 4. Independent transfer case

## Independent case: Find neighbours independently

New training rows are P at 1 cm clear, Q at 5 blocked, R at 7 blocked, S at 11 clear, T at 13 blocked. For query 6, give all distances, order the IDs by distance then alphabetically, and predict with  $k=1$  and  $k=3$ . Using the same training pool and  $k=3$ , score validation (6,blocked),(10,clear),(12,blocked). Compare the training-majority baseline.

### My independent explanation, trace and limits

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

## 5. Your learning target

A query has nearest books A and B at distance 1, with conflicting labels; C is distance 2. A program chooses whichever row occurs first. Explain the risk and define a reproducible alternative.

## 6. Solve the given case

A=[1,1] liked, B=[1,0] disliked, C=[0,0] disliked; query=[0,1].

Find nearest neighbours using differing-trait distance and  $k=1$ .

Show your trace, calculation or evidence

### Write or draw your reasoning and evidence.

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## 7. Check a changed case

Training items have one numeric feature: A=2 with label short, B=8 with label long. A new item has value 6. Use absolute distance to choose its nearest neighbour.

My independent answer and reason

**Write or draw your reasoning and evidence.**

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### Week 22 Test and reflect

## 8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

### Test record · My evidence

**Test record:** Changed input or condition

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Test record:** Expected answer or fallback

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Test record:** Observed result

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Test record:** Why the result occurred

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

## 9. Project checkpoint

Use neighboring liked books as evidence, while explaining which traits created closeness.

The project change and evidence I will keep

**Write or draw your reasoning and evidence.**

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## 10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review \_\_\_\_\_ Next step \_\_\_\_\_

## 11. Transfer practice from the original programme

A query has nearest books A and B at distance 1, with conflicting labels; C is distance 2. A program chooses whichever row occurs first. Explain the risk and define a reproducible alternative.

**My independently reasoned answer**

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Record support used: none, hint or teacher explanation.

## 12. W1 Calculate feature distances

For query 3 cm, calculate distances to all five training rows. Explain why A and B tie even though their labels differ.

**My reasoning and evidence**

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## 13. W2 Apply neighbour and tie rules

For query 3, list the selected IDs and predictions for  $k=1$  and  $k=3$  using alphabetical ID order on equal distances. Explain what changes and what remains the same.

**My reasoning and evidence**

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## 14. W3 Keep evaluation examples separate

Why must the three validation reference labels not enter the neighbour pool? Give the fixed  $k=3$  score and baseline score on the worked validation examples.

**My reasoning and evidence**

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## 15. W4 Find neighbours independently

New training rows are P at 1 cm clear, Q at 5 blocked, R at 7 blocked, S at 11 clear, T at 13 blocked. For query 6, give all distances, order the IDs by distance then alphabetically, and predict with  $k=1$  and  $k=3$ . Using the same training pool and  $k=3$ , score validation (6,blocked),(10,clear),(12,blocked). Compare the training-majority baseline.

### My reasoning and evidence

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## 16. W5 Interpret a vote and a match

Two of three neighbours vote clear. Does that prove a two-thirds chance of a correct prediction? Does an exact feature match guarantee a correct label? Explain both limits.

### My reasoning and evidence

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## 17. Feedback, return check and learning record

## Evidence · My record

**Evidence:** Worked alone / hint / model

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** First step I repaired and why

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** New case checked later; date and result

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** What I can now explain without notes

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** One remaining question

**My record:** \_\_\_\_\_

\_\_\_\_\_

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

## 18. Pictorial case practice: Nearest Neighbor Reasoning

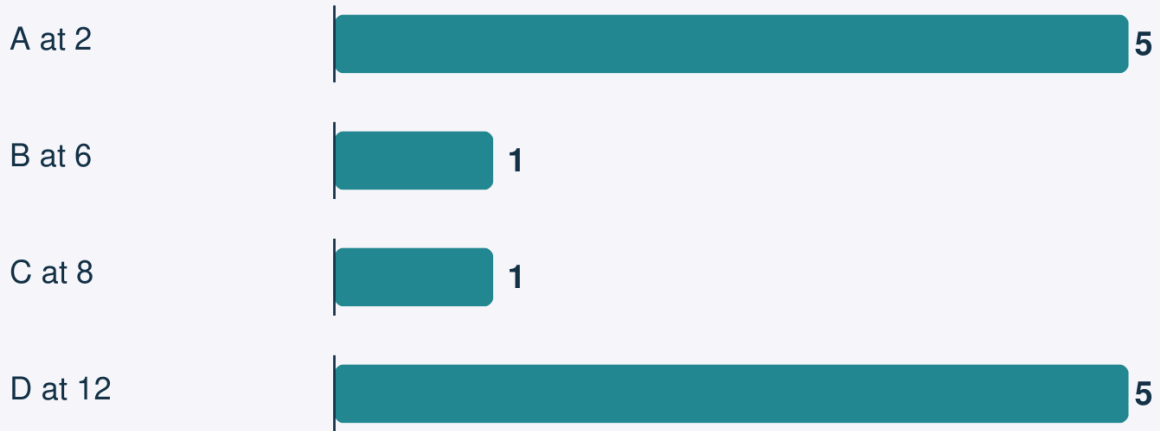
A fictional one-feature classifier predicts compact or extended book format.

Stored training rows: A at 2 units compact, B at 6 extended, C at 8 compact, D at 12 extended. Query is 7.

Distance is absolute difference; ties sort by ID; tied category vote chooses compact.

## Distances from query 7

absolute distance units



Bar length shows the supplied value; the marked baseline is zero.

## Nearest Neighbor Reasoning

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Measure distances

From 7: A=5, B=1, C=1, D=5.



#### 2. Order neighbours

Distance then ID gives B,C,A,D.



#### 3. k=1 prediction

First neighbour B is extended: predict extended.



#### 4. k=3 prediction

B,C,A vote extended,compact,compact: predict compact.

### CHANGE ONE THING

Move query from 7 to 9 while keeping training fixed.

Explain the changed result using the matching lesson.

Why do k=1 and k=3 disagree for the original query?

### My picture-based explanation and changed-case prediction

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Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Retain recommender neighbour IDs, distances, k and tie decisions so similar-book explanations can be audited.